

EC303 · MID-SEM

THE PAPERS

Three mid-sem papers — 2023, 2024 and 2025 — with every question worked out, and a **predicted paper for September 2026**. The 2023 and 2024 scans are poor in places; where a line is genuinely unreadable it is said so rather than guessed at.

WHAT ACTUALLY GETS ASKED

3 PAPERS READ END TO END

This paper has a very stable skeleton: five questions, 30 marks, and the same four areas every single year. What moves around is which corner of the DFT/FFT gets asked and how the difference equation is dressed up.

TOPIC	23	24	25	OF 3
DFT or FFT — compute, count operations, or draw the butterfly				3
Difference equation solved (step, particular, or full response)				3
z-transform, ROC, and causality / stability from it				3
Frequency response — magnitude and phase, or filter type				3
Inverse system — find it and describe its behaviour				2
A non-radix-2 FFT (radix-3, split-radix)				2
Convolution worked by hand				1
Periodicity, and energy versus power signals				1
Structure realisation (delay, adder, multiplier blocks)				1
An analog RC circuit analysed as a filter				1
DFT property applied (frequency shift, symmetry)				1
System identified from its step response				1
Time scaling / shifting inside a z-transform				1

FILLED = IT APPEARED THAT YEAR · TINTED ROWS CAME UP IN EVERY PAPER WE HAVE

READ IT THIS WAY

Four areas, every year, without exception. The DFT/FFT question, a difference equation, a z-transform with its ROC, and a frequency response. Those four are roughly 24 of the 30 marks. Everything else is the examiner's choice of garnish.

The FFT question is the one that moves. 2023 asked for a *radix-3* flow graph for $N = 9$. 2024 asked for operation counts including *split-radix*, and separately for DIT applied to the *inverse* DFT. 2025 asked for *decimation in frequency* on 16 points. So do not learn only the 8-point DIT butterfly — learn the counting formulas, both decimations, and what changes for a different radix.

Watch the ROC wording. Every year the z-transform question says “*mention the ROC in all cases*” or “*find $h(n)$ if the system is stable / causal / anti-causal*”. It is asking for **three separate answers from one algebraic $H(z)$** , and most of the marks are there rather than in the partial fractions.

2025 added something new: being handed a step response and asked to work backwards to the input. That is just differentiate-then-divide in the transform domain, and it is worth practising because it is an easy 7 marks once you see it.

PREDICTED PAPER — SEPTEMBER 2026

OUR READING, WITH FULL SOLUTIONS

Our prediction, not a leak. Built from the shape that has held for three years: five questions, 30 marks, the four guaranteed areas, and one question that works backwards through a system. Full solutions on every part.

SVNIT, Surat

Department of Electronics Engineering

Mid-Semester Examination, September 2026 — *predicted*

B.Tech. III Sem. V

DIGITAL SIGNAL PROCESSING (EC303)

TOTAL MARKS: 30

DURATION: 1 HOUR 30 MINUTES

ATTEMPT ALL QUESTIONS

Instructions. Attempt all questions. Assume necessary data if required. The figure to the right indicates marks.

- Q-1** A discrete-time LTI system is described by $y(n) - \frac{1}{4}y(n-1) = x(n) + x(n-1)$. (i) Find the transfer function and sketch the pole-zero plot. (ii) Find the magnitude and phase of the frequency response, and state whether the system is low-pass or high-pass. (iii) Find the DC gain.

07

SOLUTION

(i) Transfer function. Take the z-transform of both sides, using the delay property $x(n-k) \leftrightarrow z^{-k}X(z)$:

$\leftrightarrow z^{-k}X(z)$:

$$Y(z) - \left(\frac{1}{4}\right)z^{-1}Y(z) = X(z) + z^{-1}X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + z^{-1}}{1 - 0.25z^{-1}}$$

Multiply top and bottom by z to read the pole and zero off directly:

$$H(z) = \frac{z + 1}{z - 0.25}$$

zero at $z = -1$ pole at $z = +0.25$

Pole-zero plot: a single pole at $+0.25$ on the real axis (well inside the unit circle) and a single zero at -1 (exactly on the unit circle, at $\omega = \pi$).

```

      Im
      |
o-----x-----> Re
-1    0  0.25  1
zero      pole
(on the unit circle)

```

(ii) Frequency response. Put $z = e^{j\omega}$:

$$H(e^{j\omega}) = \frac{1 + e^{-j\omega}}{1 - 0.25e^{-j\omega}}$$

Magnitude — take the modulus of top and bottom separately:

$$\begin{aligned} |1 + e^{-j\omega}| &= \sqrt{(1+\cos \omega)^2 + \sin^2 \omega} = \sqrt{2 + 2\cos \omega} \\ |1 - 0.25 e^{-j\omega}| &= \sqrt{(1-0.25\cos \omega)^2 + (0.25 \sin \omega)^2} \\ &= \sqrt{1.0625 - 0.5 \cos \omega} \\ |H(e^{j\omega})| &= \sqrt{2 + 2\cos \omega} / \sqrt{1.0625 - 0.5 \cos \omega} \end{aligned}$$

Phase — subtract the angle of the denominator from the angle of the numerator:

$$\begin{aligned} \angle H(e^{j\omega}) &= -\arctan \left[\frac{\sin \omega}{1 + \cos \omega} \right] \\ &+ \arctan \left[\frac{0.25 \sin \omega}{1 - 0.25 \cos \omega} \right] \end{aligned}$$

Low-pass or high-pass? Evaluate at the two ends:

$$\begin{aligned} \omega = 0: \quad |H| &= \frac{\sqrt{4}}{\sqrt{0.5625}} = \frac{2}{0.75} = 2.667 \\ \omega = \pi: \quad |H| &= \frac{\sqrt{0}}{\sqrt{1.5625}} = \frac{0}{1.25} = 0 \end{aligned}$$

Low-pass — maximum gain at DC, and an exact null at $\omega = \pi$

The null is no accident: the zero sits exactly at $z = -1$, which is the point $\omega = \pi$ on the unit circle, so that frequency is killed completely.

(iii) DC gain is just $H(z)$ at $z = 1$:

$$H(1) = \frac{1+1}{1-0.25} = \frac{2}{0.75} = \frac{8}{3} = 2.667$$

THE SHORTCUT WORTH KNOWING

DC gain = $H(z)|_{z=1}$, which is the same as summing all the coefficients: $(1+1)/(1-0.25)$. And the gain at the highest frequency is $H(z)|_{z=-1}$. Two substitutions tell you the filter type without plotting anything.

- Q-2** A signal $x(n)$ has the sample values $x(0) = 1, x(1) = 2, x(2) = 3, x(3) = 4$. (i) Compute the 4-point DFT $X(k)$. (ii) Verify your answer using the property that $X(0)$ is the sum of the samples. (iii) State what $X(4), X(5), X(6)$ and $X(7)$ would be, and why. 06

SOLUTION

(i) With $N = 4$ the twiddle factor is $W_4^{nk} = e^{-j2\pi nk/4} = (-j)^{nk}$, so the whole thing is done with powers of $-j$ and no trigonometry at all.

$$X(k) = \sum_{n=0..3} x(n)(-j)^{nk}$$

$$X(0) = 1 + 2 + 3 + 4 = 10$$

$$X(1) = 1 + 2(-j) + 3(-j)^2 + 4(-j)^3$$

$$= 1 - 2j - 3 + 4j = -2 + 2j$$

$$X(2) = 1 + 2(-j)^2 + 3(-j)^4 + 4(-j)^6$$

$$= 1 - 2 + 3 - 4 = -2$$

$$X(3) = 1 + 2(-j)^3 + 3(-j)^6 + 4(-j)^9$$

$$= 1 + 2j - 3 - 4j = -2 - 2j$$

$$X(k) = \{ 10, -2+2j, -2, -2-2j \}$$

(ii) Two free checks.

- $X(0) = \sum x(n)$, because every twiddle is 1 at $k = 0$. Here $1+2+3+4 = 10$. ✓
- **Conjugate symmetry.** $x(n)$ is real, so $X(N-k)$ must be the complex conjugate of $X(k)$. Check: $X(3) = -2-2j$ is exactly the conjugate of $X(1) = -2+2j$. ✓ $X(2)$ sits on its own at $N/2$ and must therefore be purely real — it is. ✓

(iii) The DFT is **periodic in k with period N** , so $X(k+N) = X(k)$:

$$X(4) = X(0) = 10$$

$$X(5) = X(1) = -2 + 2j$$

$$X(6) = X(2) = -2$$

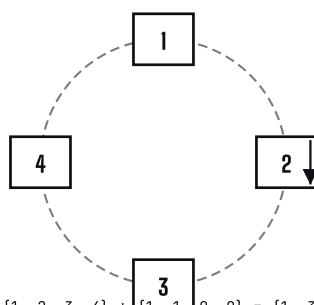
$$X(7) = X(3) = -2 - 2j$$

THE TRAP

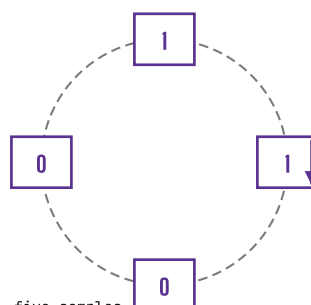
Being asked for eight values does not make it an 8-point DFT. A 4-point DFT has exactly four distinct values and then repeats forever. The 2025 paper set precisely this trap with a different set of samples.

DRAWN FOR THE ANSWER

CIRCULAR CONVOLUTION - $N = 4$
 $x(n)$



$h(-n) \bmod 4$



linear: $\{1, 2, 3, 4\} * \{1, 1, 0, 0\} = \{1, 3, 5, 7, 4\}$ — five samples

circular: the 5th wraps onto the 1st — $4 + 1 = 5$

$\{5, 3, 5, 7\}$

pad both to $3 + 2 - 1 = 5$ points first and the two agree — that is how the FFT does linear convolution.

Circular convolution, and the sample that wraps round onto the first.

- Q-3** A system has $H(z) = (3 - 4z^{-1}) / (1 - 3.5z^{-1} + 1.5z^{-2})$. Specify all possible regions of convergence, and find $h(n)$ in each case. Say which one is causal and which is stable. 05

SOLUTION

Step 1 — factor the denominator.

$$1 - 3.5z^{-1} + 1.5z^{-2} = (1 - 3z^{-1})(1 - 0.5z^{-1})$$

poles at $z = 3$ and $z = 0.5$

Step 2 — partial fractions.

$$H(z) = \frac{3 - 4z^{-1}}{[(1 - 3z^{-1})(1 - 0.5z^{-1})]}$$

$$= \frac{A}{1 - 3z^{-1}} + \frac{B}{1 - 0.5z^{-1}}$$

Multiply out and match: $3 - 4z^{-1} = A(1 - 0.5z^{-1}) + B(1 - 3z^{-1})$.

constant term : $A + B = 3$

z^{-1} term: $-0.5A - 3B = -4$

solving : $A = 2, B = 1$

$$H(z) = \frac{2}{1 - 3z^{-1}} + \frac{1}{1 - 0.5z^{-1}}$$

Step 3 — the three regions. Two poles cut the plane into three rings, and each ring is a different sequence.

ROC	H(N)	CAUSAL?	STABLE?
$ z > 3$ outside both poles	$2(3)^n u(n) + (0.5)^n u(n)$	Yes — entirely right-sided	No — the unit circle is not inside this ring
$0.5 < z < 3$ the annulus between them	$-2(3)^n u(-n-1) + (0.5)^n u(n)$	No — two-sided	Yes — this ring does contain $ z = 1$
$ z < 0.5$ inside both poles	$-2(3)^n u(-n-1) - (0.5)^n u(-n-1)$	No — entirely left-sided, anti-causal	No

The rule used twice:

$$a^n u(n) \leftrightarrow \frac{1}{1 - az^{-1}}, \text{ROC } |z| > |a| \text{ (right - sided)}$$

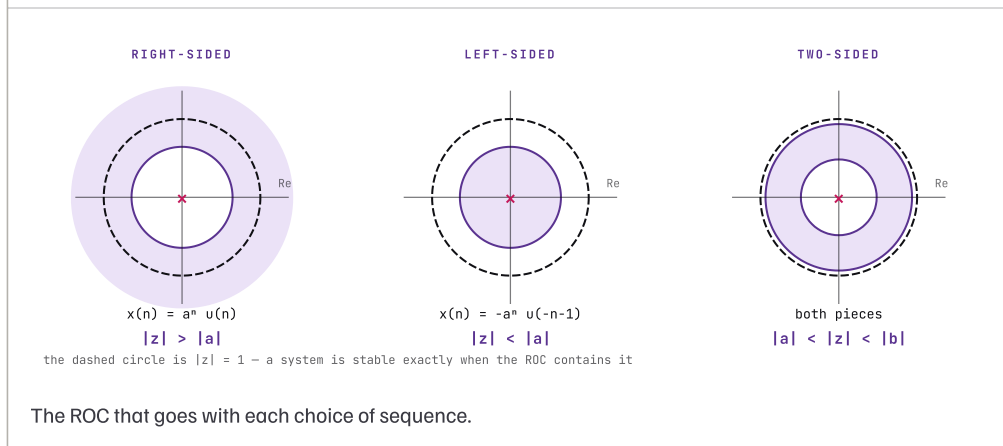
$$-a^n u(-n-1) \leftrightarrow \frac{1}{1 - az^{-1}}, \text{ROC } |z| < |a| \text{ (left - sided)}$$

Same algebra, opposite sequence — which is exactly why the ROC must always be stated.

THE CONCLUSION TO WRITE OUT

This system **cannot be both causal and stable**. Causal means the ROC is outside the outermost pole, $|z| > 3$, which excludes the unit circle. Stable means the ROC contains the unit circle, which forces the annulus, which is two-sided. A causal system is stable only when *every* pole lies inside the unit circle, and the pole at $z = 3$ does not.

DRAWN FOR THE ANSWER



Q-4 Explain the decimation-in-time radix-2 FFT algorithm for an 8-point DFT. Draw the butterfly diagram, state the required input ordering, and count the complex multiplications and additions. Compare with the direct DFT. 06

SOLUTION

The idea. Split $x(n)$ into its even-indexed and odd-indexed samples. Each half is an $N/2$ -point DFT, and the two are recombined with one twiddle multiplication per output:

$$\begin{aligned} X(k) &= G(k) + W_N^k H(k) \\ X(k + N/2) &= G(k) - W_N^k H(k) \quad \text{for } k = 0 \dots N/2 - 1 \\ G(k) &= \text{DFT of the even samples, } H(k) = \text{DFT of the odd samples} \end{aligned}$$

Repeat the split until only 2-point DFTs are left. For $N = 8$ that takes $\log_2 8 = 3$ stages, each with $N/2 = 4$ butterflies.

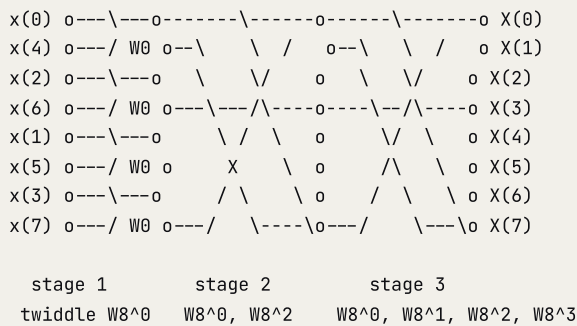
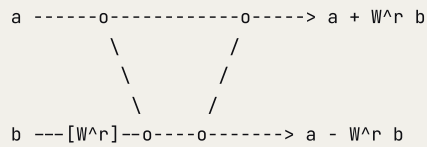
Input ordering — bit reversal. Splitting into evens and odds repeatedly leaves the input in bit-reversed order. Write each index in 3 bits and reverse them:

$$\begin{aligned} 0 &= 000 \rightarrow 000 = 0 \cdot 4 = 100 \rightarrow 001 = 1 \\ 1 &= 001 \rightarrow 100 = 4 \cdot 5 = 101 \rightarrow 101 = 5 \\ 2 &= 010 \rightarrow 010 = 2 \cdot 6 = 110 \rightarrow 011 = 3 \\ 3 &= 011 \rightarrow 110 = 6 \cdot 7 = 111 \rightarrow 111 = 7 \end{aligned}$$

input order : $x(0)x(4)x(2)x(6)x(1)x(5)x(3)x(7)$
 output order : $X(0)X(1)X(2)X(3)X(4)X(5)X(6)X(7)$

natural

The butterfly. Each one takes two values, multiplies the lower by a twiddle, then adds and subtracts:



Operation count.

$$\text{complex multiplications} = \left(\frac{N}{2}\right) \log_2 N = 4 \times 3 = 12$$

$$\text{complex additions} = N \log_2 N = 8 \times 3 = 24$$

direct DFT:

$$\text{complex multiplications} = N^2 = 64$$

$$\text{complex additions} = N(N - 1) = 56$$

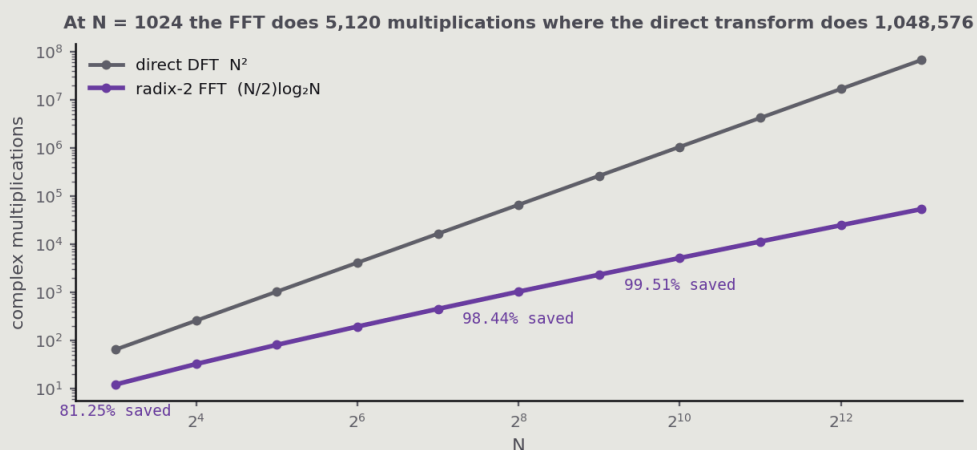
12 multiplications instead of 64 – about 5.3 times fewer

SAY THIS ABOUT THE SAVING

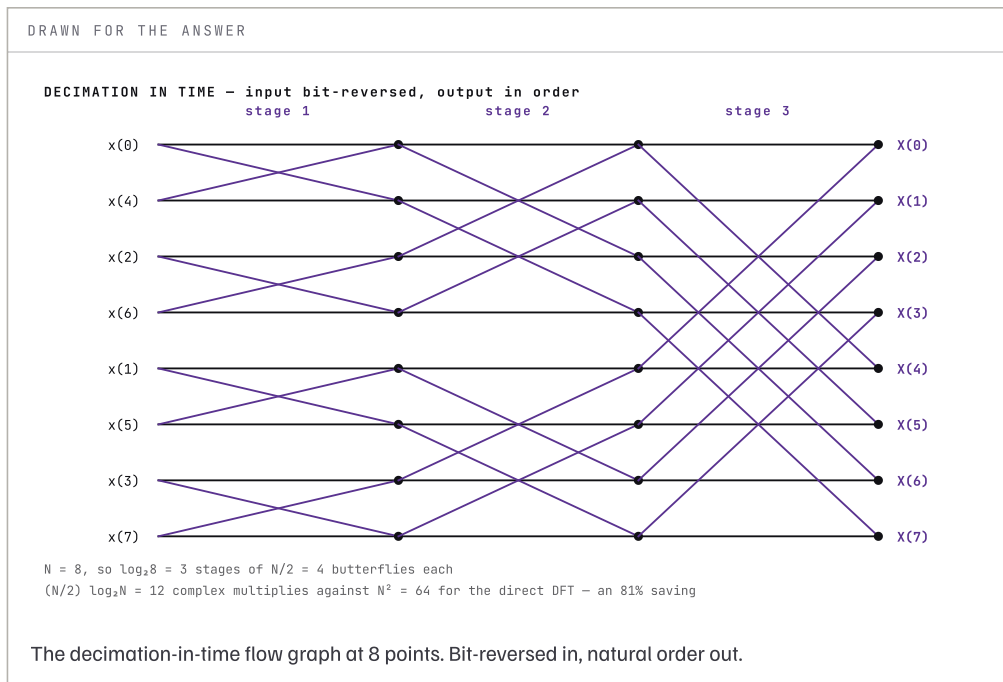
The saving grows with N, and fast: at N = 8 it is about 5×, at N = 512 it is 114×, and at N = 1024 it is 205×. That is why the algorithm was worth inventing — it turns an N^2 problem into an $N \log N$ one.

Also worth a mark: the computation is **in place**. Each butterfly writes its two results back over the two inputs it just read, so no extra array is needed; bit reversal is the price paid for that.

COMPUTED · TOOLS/MAKE-FIGURES.PY



Direct evaluation against the radix-2 FFT, with the saving marked at three values of N.



Q-5

A system is described by $y(n) - \frac{1}{2}y(n-1) = x(n)$, and the input is $x(n) = (\frac{1}{4})^n u(n)$ with the system initially relaxed. Find the complete response using the homogeneous-plus-particular method, and verify it with the z-transform.

06

SOLUTION

Method 1 — homogeneous plus particular.

Homogeneous solution. Set the right-hand side to zero and try $y_h(n) = C\lambda^n$:

$$\lambda^n - \left(\frac{1}{2}\right) \lambda^{n-1} = 0 \rightarrow \lambda = \frac{1}{2}$$

$$y_h(n) = C \left(\frac{1}{2}\right)^n$$

Particular solution. The forcing function is $(\frac{1}{4})^n$, so try $y_p(n) = K(\frac{1}{4})^n$ for $n \geq 0$:

$$K\left(\frac{1}{4}\right)^n - \left(\frac{1}{2}\right) K\left(\frac{1}{4}\right)^{n-1} = \left(\frac{1}{4}\right)^n$$

divide through by $(\frac{1}{4})^n$:

$$K - \left(\frac{1}{2}\right) K \left(\frac{1}{4}\right) = 1$$

$$K - 2K = 1$$

$$-K = 1 \rightarrow K = -1$$

$$y_p(n) = -\left(\frac{1}{4}\right)^n$$

Complete solution.

$$y(n) = C \left(\frac{1}{2}\right)^n - \left(\frac{1}{4}\right)^n$$

Apply the initial condition. Relaxed means $y(-1) = 0$, so from the difference equation at $n = 0$:

$$y(0) = \frac{1}{2}y(-1) + x(0) = 0 + 1 = 1.$$

$$y(0) = C - 1 = 1 \rightarrow C = 2$$

$$y(n) = 2\left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n, \text{ for } n \geq 0$$

Method 2 — the z-transform, as a check.

$$\begin{aligned} H(z) &= \frac{1}{1 - 0.5z^{-1}} \\ X(z) &= \frac{1}{1 - 0.25z^{-1}} \\ Y(z) &= \frac{1}{[(1 - 0.5z^{-1})(1 - 0.25z^{-1})]} \\ &= \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.25z^{-1}} \end{aligned}$$

$$1 = A(1 - 0.25z^{-1}) + B(1 - 0.5z^{-1})$$

$$\text{constant: } A + B = 1$$

$$z^{-1}: -0.25A - 0.5B = 0 \rightarrow A = -2B$$

$$-2B + B = 1 \rightarrow B = -1, A = 2$$

$$\begin{aligned} Y(z) &= \frac{2}{1 - 0.5z^{-1}} - \frac{1}{1 - 0.25z^{-1}} \\ y(n) &= 2(0.5)^n u(n) - (0.25)^n u(n) \quad \text{same answer} \end{aligned}$$

Check the first two samples directly from the recursion $y(n) = \frac{1}{2}y(n-1) + x(n)$:

$$y(0) = 0 + 1 = 1 \text{ formula: } 2(1) - 1 = 1 \text{ OK}$$

$$y(1) = 0.5 + 0.25 = 0.75 \text{ formula: } 2(0.5) - 0.25 = 0.75 \text{ OK}$$

$$y(2) = 0.375 + 0.0625 = 0.4375 \text{ formula: } 2(0.25) - 0.0625 = 0.4375 \text{ OK}$$

WHICH METHOD TO USE IN THE EXAM

If the question says “using homogeneous and particular solution”, you must show both parts — the z-transform alone will not score. But do the z-transform anyway on rough paper: it takes two minutes and catches arithmetic slips, which is where these marks are actually lost.

MID-SEM, SEPTEMBER 2025

EC303 · 30 MARKS · 23 SEPT 2025

A clean scan and a very typical paper: five questions, the four standing areas, plus one working-backwards question that was new that year.

SVNIT, Surat

Department of Electronics Engineering
Mid-Semester Examination, September 2025
B.Tech. III Sem. V · 23/09/2025

DIGITAL SIGNAL PROCESSING (EC303)

TOTAL MARKS: 30

DURATION: 1 HOUR 30 MINUTES

FIVE QUESTIONS

- Q-1** A discrete-time LTI system is described by $y(n) - \frac{1}{2}y(n-1) = x(n)$, where $x(n) = \{1, 2, 3, 4\}$ for $0 \leq n \leq 3$. (i) Calculate the magnitude and phase of the transfer function. (ii) Compute the magnitude and phase of the output signal $y(n)$. 07

SOLUTION

(i) Transfer function.

$$Y(z) - 0.5z^{-1}Y(z) = X(z)$$

$$H(z) = \frac{1}{1 - 0.5z^{-1}} \text{ pole at } z = 0.5, \text{ zero at } z = 0$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega}}$$

Magnitude. Write the denominator in rectangular form: $1 - 0.5\cos\omega + j(0.5\sin\omega)$.

$$\begin{aligned} |1 - 0.5e^{-j\omega}| &= \sqrt{(1 - 0.5\cos\omega)^2 + (0.5\sin\omega)^2} \\ &= \sqrt{1 - \cos\omega + 0.25\cos^2\omega + 0.25\sin^2\omega} \\ &= \sqrt{1.25 - \cos\omega} \\ |H(e^{j\omega})| &= \frac{1}{\sqrt{1.25 - \cos\omega}} \end{aligned}$$

Phase.

$$\angle H(e^{j\omega}) = -\arctan \left[\frac{0.5\sin\omega}{1 - 0.5\cos\omega} \right]$$

Check the two ends:

$$\begin{aligned} \omega = 0: \quad |H| &= \frac{1}{\sqrt{0.25}} = 2 \text{ phase} = 0 \\ \omega = \pi: \quad |H| &= \frac{1}{\sqrt{2.25}} = 0.667 \text{ phase} = 0 \end{aligned}$$

Gain 2 falling to 0.667 — a **low-pass** system with a DC gain of 2.

(ii) The output. The input is a 4-point finite sequence, so

$$X(e^{j\omega}) = 1 + 2e^{-j\omega} + 3e^{-j2\omega} + 4e^{-j3\omega}$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

$$|Y(e^{j\omega})| = |H| \times |X| = \frac{|X(e^{j\omega})|}{\sqrt{1.25 - \cos \omega}}$$

$$\angle Y(e^{j\omega}) = \angle H + \angle X$$

Because multiplying in the frequency domain means **magnitudes multiply and phases add**, that is the complete answer in closed form.

The samples themselves are easier to get straight from the recursion $y(n) = \frac{1}{2}y(n-1) + x(n)$, with $y(-1) = 0$:

$$y(0) = 0 + 1 = 1$$

$$y(1) = 0.5 + 2 = 2.5$$

$$y(2) = 1.25 + 3 = 4.25$$

$$y(3) = 2.125 + 4 = 6.125$$

$$y(4) = 3.0625 + 0 = 3.0625$$

input has ended; it now just decays

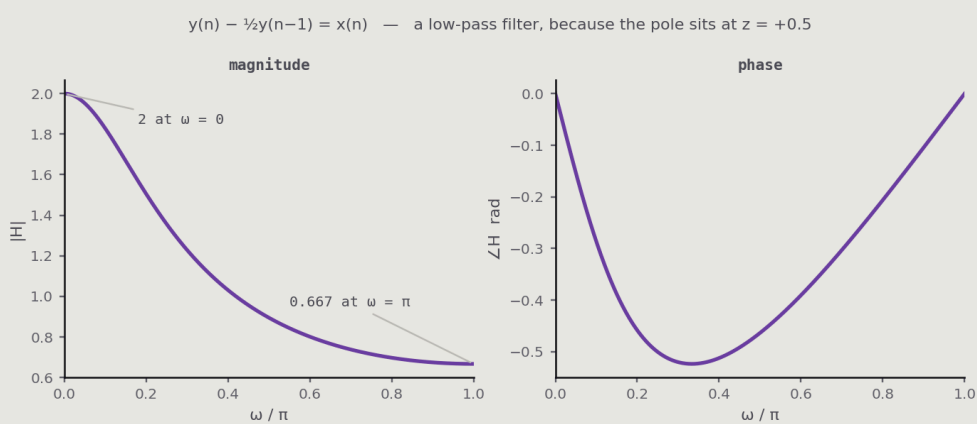
$$y(5) = 1.53125, y(6) = 0.765625, \dots$$

So the output is **not** a 4-point sequence — the pole at 0.5 gives it an infinite decaying tail. Say that; it is worth a mark.

THE IDENTITY THAT SAVES TIME

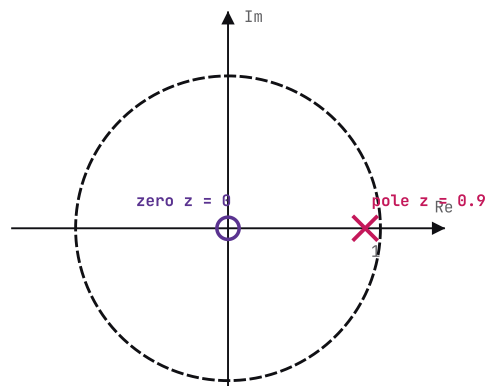
$|1 - ae^{-j\omega}|^2 = 1 - 2a\cos\omega + a^2$. With $a = 0.5$ that is $1.25 - \cos\omega$ immediately, without expanding $\sin^2$ and $\cos^2$ separately. It comes up in every single one of these questions.

COMPUTED · TOOLS/MAKE-FIGURES.PY



$|H(\omega)|$ and $\angle H(\omega)$ evaluated on 800 points. The pole at $z = +0.5$ is what makes it low-pass, and the two marked values are the ones the answer computes by hand.

DRAWN FOR THE ANSWER



$$H(z) = b z / (z - 0.9) \text{ ROC } |z| > 0.9$$

the pole is inside the unit circle, so the filter is stable

it sits near $z = +1$, which is $\omega = 0$ — so this is a low-pass filter

Where the pole sits decides everything: inside the circle means stable, and near $z = +1$ means low-pass.

- Q-2** A signal $x(n]$ has sample values $x(0) = -2, x(1) = 1, x(2) = 1$ and $x(3) = 2$. The DFT of the signal is $X(k)$. Calculate $X(0), X(1), X(2), X(3), X(4), X(5), X(6)$ and $X(7)$. 06

SOLUTION

Four samples means a **4-point** DFT, so $W_4^{nk} = (-j)^{nk}$.

$$X(0) = -2 + 1 + 1 + 2 = 2$$

$$\begin{aligned} X(1) &= -2 + 1(-j) + 1(-j)^2 + 2(-j)^3 \\ &= -2 - j - 1 + 2j = -3 + j \end{aligned}$$

$$\begin{aligned} X(2) &= -2 + 1(-j)^2 + 1(-j)^4 + 2(-j)^6 \\ &= -2 - 1 + 1 - 2 = -4 \end{aligned}$$

$$\begin{aligned} X(3) &= -2 + 1(-j)^3 + 1(-j)^6 + 2(-j)^9 \\ &= -2 + j - 1 - 2j = -3 - j \end{aligned}$$

The remaining four are the trap. The DFT is periodic in k with period $N = 4$, so it simply repeats:

$$X(4) = X(0) = 2$$

$$X(5) = X(1) = -3 + j$$

$$X(6) = X(2) = -4$$

$$X(7) = X(3) = -3 - j$$

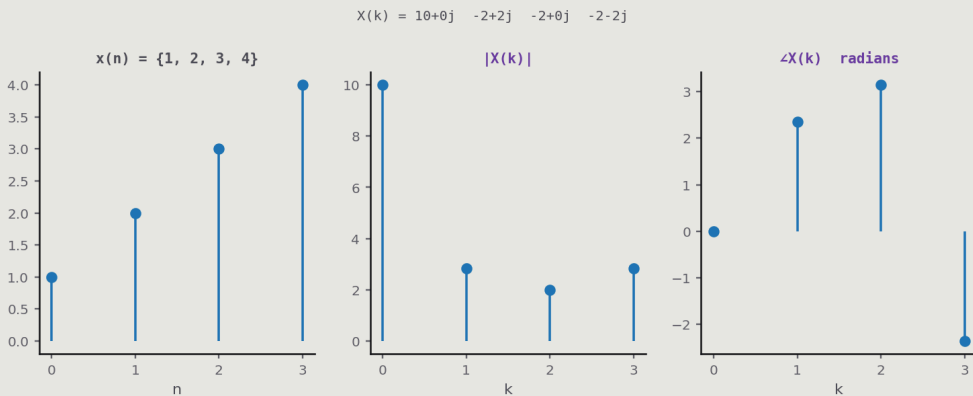
$$X = \{ 2, -3+j, -4, -3-j, 2, -3+j, -4, -3-j \}$$

Two checks. $X(0) = \sum x(n) = 2 \checkmark$. And $x(n]$ is real, so $X(3)$ must be the conjugate of $X(1)$: $-3-j$ against $-3+j \checkmark$. $X(2)$ sits at $N/2$ and must be purely real — it is.

DO NOT BE TRICKED INTO AN 8-POINT DFT

Being asked for eight values does not change N . If the examiner had wanted an 8-point DFT they would have had to give eight samples, or say “zero-pad to 8” — and that would give genuinely different numbers, not repeats.

COMPUTED · TOOLS/MAKE-FIGURES.PY



The same four samples through `np.fft.fft`, with the magnitude and the phase. Every number in the table above comes back out of the array.

Q-3 A signal $x(n)$ is given by $x(n) = a^n \cdot u(n+2)$, and another signal $y(n)$ is related to $x(n)$ by $y(n) = x(\frac{1}{2}n + 2)$. Calculate the Z-transform of $y(n)$. Also mention the ROC in all cases. 05

SOLUTION

Step 1 — $X(z)$. The step $u(n+2)$ means $x(n)$ is non-zero from $n = -2$ upward.

$$\begin{aligned}
 X(z) &= \sum_{n=-2}^{\infty} a^n z^{-n} \\
 &\text{substitute } n = m - 2, \text{ so } m \text{ runs from } 0: \\
 &= \sum_{m=0}^{\infty} a^{(m-2)} z^{-(m-2)} \\
 &= a^{-2} z^2 \sum_{m=0}^{\infty} (a z^{-1})^m \\
 &= a^{-2} z^2 / (1 - a z^{-1})
 \end{aligned}$$

ROC of $X(z)$: the geometric series needs $|a \cdot z^{-1}| < 1$, so $|z| > |a|$. It is right-sided, so the ROC is the outside of the circle through the pole. Because the sequence starts at $n = -2$ there is a z^2 term, so $z = \infty$ is excluded.

Step 2 — what $y(n) = x(\frac{1}{2}n + 2)$ means. The argument must be a whole number, so $y(n)$ is defined for **even n only** and is taken as zero at odd n . That is an **upsampling by 2** combined with a **time advance of 2**.

$$\begin{aligned}
 y(n) &= x\left(\frac{n}{2} + 2\right) \\
 &\text{for } n \text{ even, } 0 \text{ for } n \text{ odd}
 \end{aligned}$$

Step 3 — the two properties needed.

$$\begin{aligned}
 \text{time advance : } x(n+k) &\longleftrightarrow z^k X(z) \\
 \text{upsampling : } v(n/2) \text{ (0 odd)} &\longleftrightarrow V(z^2), \text{ ROC: } |z| > \sqrt{R}
 \end{aligned}$$

Apply them in that order. Let $v(n) = x(n+2)$, so $V(z) = z^2 X(z)$. Then y is v upsampled:

$$\begin{aligned}
 Y(z) &= V(z^2) = (z^2)^2 X(z^2) = z^4 X(z^2) \\
 &= \frac{z^4 \cdot a^{-2} (z^2)^2}{1 - az^{-2}} \\
 Y(z) &= \frac{a^{-2} z^8}{1 - az^{-2}}
 \end{aligned}$$

ROC of Y(z): replacing z by z^2 squares the boundary radius, so $|z^2| > |a|$ becomes

$$|z| > \sqrt{|a|} \quad \text{and} \quad z = \infty \text{ excluded}$$

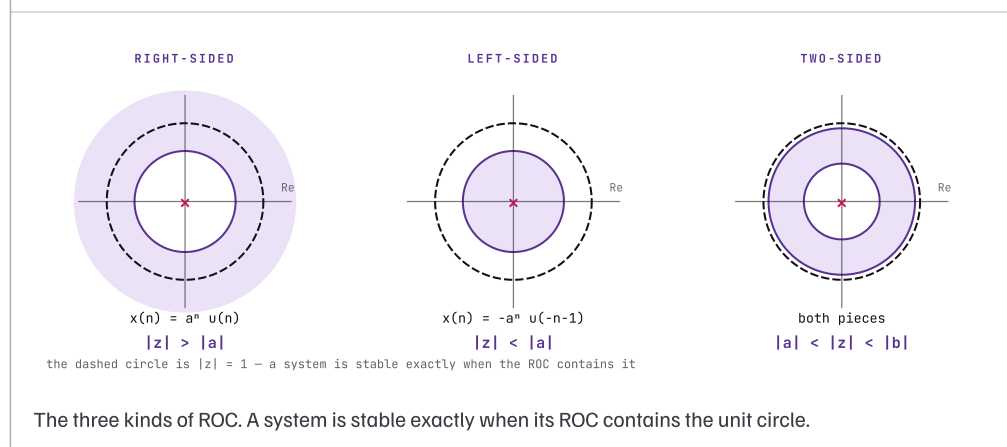
“In all cases” — the three readings the examiner is checking:

CASE	ROC OF X(Z)	ROC OF Y(Z)	STABLE?
$ a < 1$	$ z > a $	$ z > \sqrt{ a }$	Yes — the ROC contains the unit circle
$ a = 1$	$ z > 1$	$ z > 1$	No — the pole sits <i>on</i> the unit circle
$ a > 1$	$ z > a $	$ z > \sqrt{ a }$	No — the unit circle is outside the ROC

BE EXPLICIT ABOUT THE READING

$x(\frac{1}{2}n + 2)$ is genuinely ambiguous in discrete time, because $\frac{1}{2}n$ is not an integer for odd n . State your interpretation in one line before you start — “defined for even n , zero elsewhere, i.e. upsampling by 2” — and the examiner can follow you. Answering without saying which reading you took is how marks get lost here.

DRAWN FOR THE ANSWER



Q-4

Explain the Decimation-in-frequency radix-2 FFT algorithm for a 16-point DFT calculation. Also draw the butterfly diagram with a detailed explanation of each step.

05

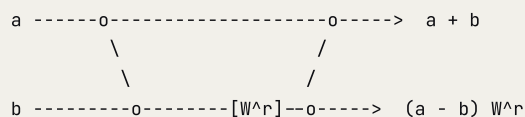
SOLUTION

How DIF splits. Where DIT splits the *input* into evens and odds, DIF splits the input into its **first half and second half**, and splits the *output* into even and odd frequency bins.

$$\begin{aligned}
 X(2r) &= \text{DFT}_{\{N/2\}} \text{ of } [x(n) + x(n + N/2)] \\
 X(2r+1) &= \text{DFT}_{\{N/2\}} \text{ of } [(x(n) - x(n + N/2)) W_N^n] \\
 \text{for } n &= 0 \dots N/2 - 1
 \end{aligned}$$

So the first operation is an add and a subtract on samples $N/2$ apart, and the twiddle multiplication comes **after** the subtraction — the opposite order to DIT.

The DIF butterfly:



For $N = 16$:

```
stages          = log2(16) = 4
butterflies/stage = N/2    = 8
total butterflies = 8 x 4 = 32
```

stage 1: pairs 16 apart? no - pairs $N/2 = 8$ apart:

$x(0) \& x(8), x(1) \& x(9), \dots x(7) \& x(15)$ twiddles $W^{16^0} \dots W^{16^7}$

stage 2: pairs 4 apart within each half twiddles $W^{16^0, 2, 4, 6}$

stage 3: pairs 2 apart within each quarter twiddles $W^{16^0, 4}$

stage 4: pairs 1 apart twiddle W^{16^0} only

```
x(0) o--\-----o---- ... ----o X(0)
x(1) o--\\-----o---- ... ----o X(8)
      :      (add / subtract      :      :
x(7) o--\\ across 8 positions)   :      :
x(8) o--//---[W16^0]-----o---- ... ----o X(4)
x(9) o--//---[W16^1]-----o---- ... ----o X(12)
      :      :                  :      :
x(15) o--/----[W16^7]-----o---- ... ----o X(15)
```

stage 1	stage 2	stage 3	stage 4
$N/2$ apart	$N/4$ apart	$N/8$ apart	adjacent

Ordering. Input in **natural order**, output in **bit-reversed order** — the exact mirror of DIT. For $N = 16$ the output sequence is

0, 8, 4, 12, 2, 10, 6, 14, 1, 9, 5, 13, 3, 11, 7, 15

Operation count for $N = 16$:

$$\text{complex multiplications} = \left(\frac{N}{2}\right) \log_2 N = 8 \times 4 = 32$$

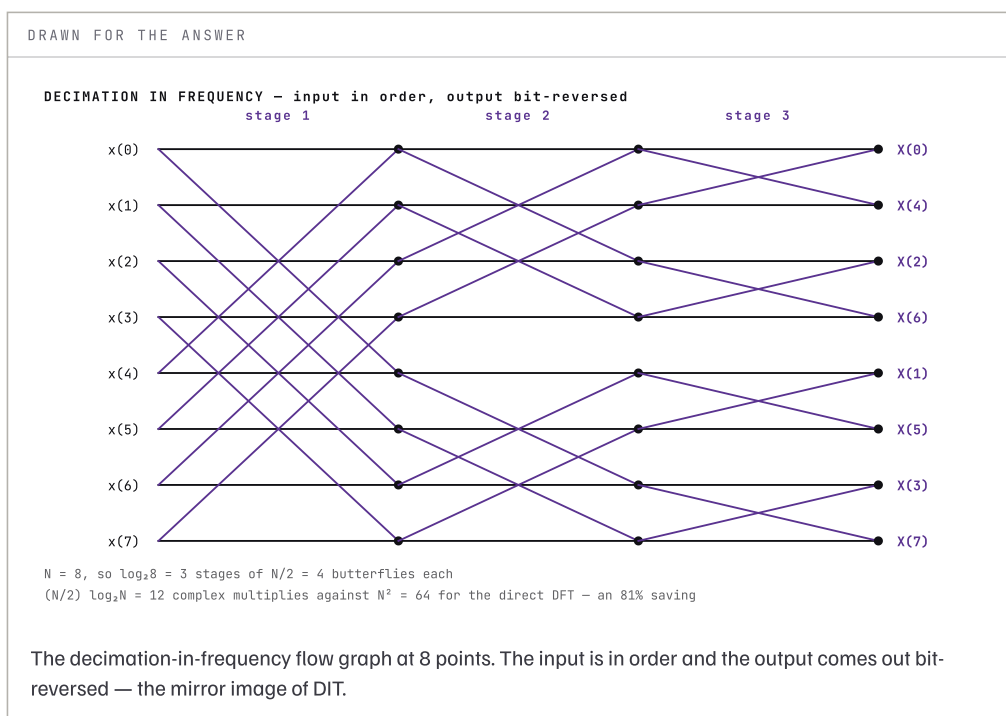
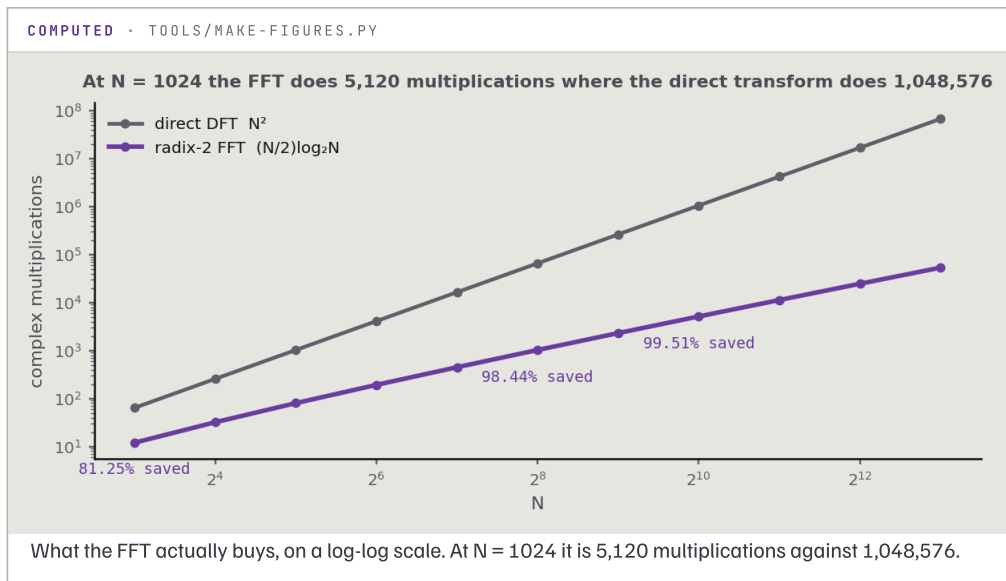
$$\text{complex additions} = N \log_2 N = 16 \times 4 = 64$$

$$\text{direct DFT : } 16^2 = 256 \text{ multiplications, } 16 \times 15 = 240 \text{ additions}$$

$$\text{saving : } \frac{256}{32} = 8 \text{ times fewer multiplications}$$

DIT OR DIF – WHICH TO WRITE

They cost **exactly the same**. The only differences are (a) which end is bit-reversed and (b) whether the twiddle multiply comes before or after the add-subtract. If a question asks you to compare them, those two sentences plus the identical operation counts are the whole answer.



Q-5

A pressure gauge that can be modelled as an LTI system has an output $(1 - e^{-t} - te^{-t})u(t)$ for a unit step input. For a certain input $x(t)$ the output observed is $(2 - 3e^{-t} + e^{-3t})u(t)$. What should the input be? Calculate it.

07

SOLUTION

Step 1 — get the impulse response from the step response. The impulse response is the derivative of the step response (equivalently, multiply by s in the transform domain).

$$s(t) = (1 - e^{-t} - te^{-t})u(t)$$

$$S(s) = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}$$

$$H(s) = s \cdot S(s) = 1 - \frac{s}{s+1} - \frac{s}{(s+1)^2}$$

Combine over $(s+1)^2$:

$$\begin{aligned} H(s) &= \frac{[(s+1)^2 - s(s+1) - s]}{(s+1)^2} \\ &= \frac{[s^2 + 2s + 1 - s^2 - s - s]}{(s+1)^2} \\ &= \frac{1}{(s+1)^2} \end{aligned}$$

$$H(s) = 1/(s+1)^2, \text{ so } h(t) = t \cdot e^{-t} u(t)$$

Check it the other way: differentiating $(1 - e^{-t} - te^{-t})$ gives $e^{-t} - (e^{-t} - te^{-t}) = te^{-t}$. ✓

Step 2 — transform the observed output.

$$\begin{aligned} y(t) &= (2 - 3e^{-t} + e^{-3t})u(t) \\ Y(s) &= \frac{2}{s} - \frac{3}{s+1} + \frac{1}{s+3} \end{aligned}$$

Put it over the common denominator $s(s+1)(s+3)$ — the numerator collapses beautifully:

$$\begin{aligned} \frac{2(s+1)(s+3) - 3s(s+3) + s(s+1)}{[s(s+1)(s+3)]} &= \frac{2(s^2 + 4s + 3) - 3(s^2 + 3s) + (s^2 + s)}{[s(s+1)(s+3)]} \\ &= \frac{2s^2 + 8s + 6 - 3s^2 - 9s + s^2 + s}{[s(s+1)(s+3)]} \\ &= \frac{(2 - 3 + 1)s^2 + (8 - 9 + 1)s + 6}{[s(s+1)(s+3)]} \\ &= \frac{6}{[s(s+1)(s+3)]} \\ Y(s) &= \frac{6}{[s(s+1)(s+3)]} \end{aligned}$$

Step 3 — divide. Since $Y(s) = X(s)H(s)$, the input is $X(s) = Y(s)/H(s) = Y(s) \cdot (s+1)^2$:

$$\begin{aligned} X(s) &= \frac{6(s+1)^2}{[s(s+1)(s+3)]} \\ &= \frac{6(s+1)}{[s(s+3)]} \end{aligned}$$

Step 4 — partial fractions and invert.

$$\begin{aligned} \frac{6(s+1)}{[s(s+3)]} &= \frac{A}{s} + \frac{B}{s+3} \\ 6(s+1) &= A(s+3) + Bs \\ s=0: \quad 6 &= 3A \rightarrow A=2 \\ s=-3: \quad 6(-2) &= -3B \rightarrow B=4 \\ X(s) &= \frac{2}{s} + \frac{4}{s+3} \end{aligned}$$

$$x(t) = (2 + 4e^{-3t}) u(t)$$

Verification — push it back through the system:

$$\begin{aligned}
 Y(s) &= X(s)H(s) = \frac{\left[\frac{2}{s} + \frac{4}{s+3}\right]}{(s+1)^2} \\
 &= \frac{[2(s+3) + 4s]}{[s(s+3)(s+1)^2]} \\
 &= \frac{6s+6}{[s(s+3)(s+1)^2]} \\
 &= \frac{6(s+1)}{[s(s+3)(s+1)^2]} \\
 &= \frac{6}{[s(s+1)(s+3)]} \quad \text{matches Step 2}
 \end{aligned}$$

THE SHAPE OF THIS QUESTION

It is deconvolution, and it is only three moves: **differentiate the step response to get h**, **transform both**, **divide**. The algebra looks heavy but almost always cancels — here $(s+1)^2$ against $(s+1)$ — because the examiner built the question backwards from a tidy answer. If nothing cancels, check your arithmetic before pressing on.

A four-question paper. Parts of the scan are badly degraded; the questions below are given as far as they are legible, and where a coefficient genuinely cannot be read it is said so and the method is worked on a representative version instead.

SVNIT, Surat

Department of Electronics Engineering

Mid-Semester Examination, September 2024

B.Tech. III Sem. V · 23/09/2024

DIGITAL SIGNAL PROCESSING

TOTAL MARKS: 30

DURATION: 1 HOUR 30 MINUTES

ATTEMPT ALL QUESTIONS

Q-1 a

The z-transform $H(z)$ of a system has three poles. Comment on the causality and stability of the system.
(The pole values are not legible in the scan; the method below applies to any of them.)

03

SOLUTION

The two tests, and they are separate.

PROPERTY	TEST ON THE ROC	TEST ON THE POLES
Causal	ROC is the <i>outside</i> of the outermost pole, and $H(z)$ has no positive powers of z	—
Stable	ROC <i>contains</i> the unit circle $ z = 1$	—
Both at once	—	Every pole strictly inside the unit circle

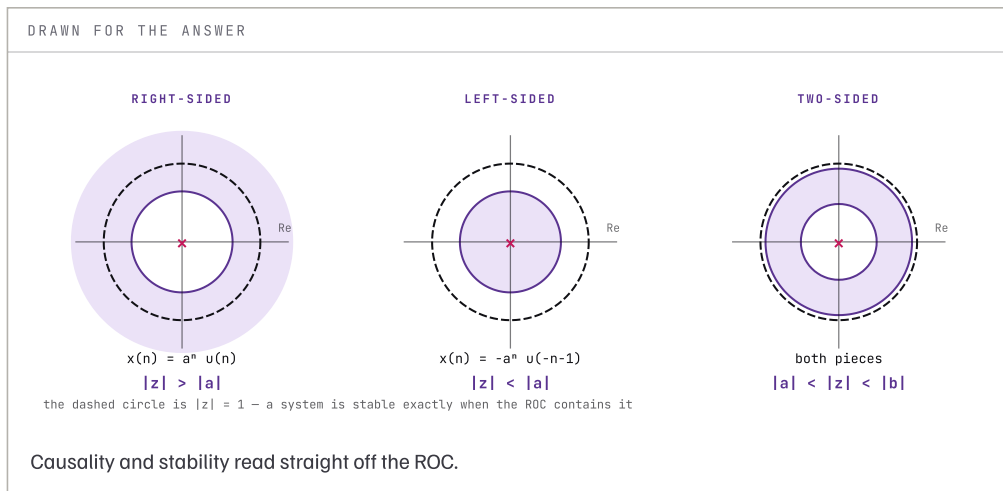
Worked on a representative system with poles at 0.5, 2 and 5:

$$H(z) = 1 / [(1 - 0.5z^{-1})(1 - 2z^{-1})(1 - 5z^{-1})]$$

three poles → four possible ROCs:

$ z > 5$	causal	NOT stable	(unit circle outside)
$2 < z < 5$	two-sided	NOT stable	
$0.5 < z < 2$	two-sided	STABLE	(contains $ z = 1$)
$ z < 0.5$	anti-causal	NOT stable	

Conclusion: with poles outside the unit circle the system **cannot be both causal and stable**. Only one ROC is stable, and it is a two-sided one. State which ROC you are assuming before answering — the question is meaningless without it.



Q-1 b Determine the unit step response for the system described by the second-order difference equation $y(n] - 4y(n-1) + 4y(n-2) = x(n) - x(n-1)$, with the system initially at rest.

04

SOLUTION

Step 1 — transfer function.

$$H(z) = \frac{1 - z^{-1}}{1 - 4z^{-1} + 4z^{-2}} = \frac{1 - z^{-1}}{(1 - 2z^{-1})^2}$$

Note the denominator is a perfect square — a **double pole at $z = 2$** .

Step 2 — the input. For a unit step, $X(z) = 1/(1 - z^{-1})$.

Step 3 — and here is the gift:

$$Y(z) = H(z) X(z) = (1 - z^{-1}) / (1 - 2z^{-1})^2 = 1 / (1 - z^{-1})$$

the $(1 - z^{-1})$ cancels exactly:

$$Y(z) = 1 / (1 - 2z^{-1})^2$$

Step 4 — invert. Use the standard pair

$$(n+1)a^n u(n) \leftrightarrow \frac{1}{(1 - az^{-1})^2}$$

$$y(n) = (n+1) \cdot 2^n \cdot u(n)$$

Check the first few against the recursion $y(n) = 4y(n-1) - 4y(n-2) + x(n) - x(n-1)$, with $y(-1) = y(-2) = 0$ and $x = u(n)$:

$$\begin{aligned}
 n = 0: \quad y(0) &= 0 - 0 + 1 - 0 = 1 \text{ formula: } (1)(1) = 1 \text{ OK} \\
 n = 1: \quad y(1) &= 4(1) - 0 + 1 - 1 = 4 \text{ formula: } (2)(2) = 4 \text{ OK} \\
 n = 2: \quad y(2) &= 4(4) - 4(1) + 0 = 12 \text{ formula: } (3)(4) = 12 \text{ OK} \\
 n = 3: \quad y(3) &= 4(12) - 4(4) = 32 \text{ formula: } (4)(8) = 32 \text{ OK}
 \end{aligned}$$

Comment: the step response grows without limit, because the double pole at $z = 2$ is outside the unit circle. The system is **unstable** — a bounded input has produced an unbounded output.

Q-1 c The DFT of a 5-point signal $x(n)$, $0 \leq n \leq 4$, is $X(k) = \{5, 6, 1, 2, 9\}$. A new signal $g(n)$ is defined by $g(n) = W_5^{2n} \cdot x(n)$ for $0 \leq n \leq 4$. What are the DFT coefficients $G(k)$?

03

SOLUTION

Recognise the property. Multiplying by a complex exponential in time is a **shift in frequency**:

$$x(n) W_N^{k_0 n} \longleftrightarrow X((k + k_0) \bmod N)$$

Here $k_0 = 2$ and $N = 5$, so every coefficient moves two places — circularly.

$$\begin{aligned}
 G(k) &= X((k + 2) \bmod 5) \\
 G(0) &= X(2) = 1 \\
 G(1) &= X(3) = 2 \\
 G(2) &= X(4) = 9 \\
 G(3) &= X(0) = 5 & 5 \bmod 5 = 0 \\
 G(4) &= X(1) = 6 & 6 \bmod 5 = 1
 \end{aligned}$$

$$G(k) = \{ 1, 2, 9, 5, 6 \}$$

Check: the set of values is unchanged — only their order has rotated. That must be so, since multiplying by a unit-magnitude exponential cannot change any magnitude.

WHICH WAY DOES IT ROTATE?

Work one term from the definition rather than trusting memory. $G(k) = \sum x(n) W_N^{2n} W_N^{nk} = \sum x(n) W_N^{n(k+2)} = X(k+2)$. The sign of k_0 in the exponent tells you the direction, and getting it backwards is the only real way to lose these marks.

Q-2 a Consider a 2^8 -point DFT. How many computations are required if the direct DFT algorithm is used? What percentage improvement in computation is obtained if the radix-2 FFT algorithm is used? Also find the percentage improvement if the same DFT is done using the split-radix algorithm.

03

SOLUTION

$$N = 2^8 = 256.$$

Direct DFT:

$$\begin{aligned}\text{complex multiplications} &= N^2 = 256^2 = 65,536 \\ \text{complex additions} &= N(N-1) = 256 \times 255 = 65,280\end{aligned}$$

Radix-2 FFT:

$$\begin{aligned}\text{complex multiplications} &= (N/2) \log_2 N = 128 \times 8 = 1,024 \\ \text{complex additions} &= N \log_2 N = 256 \times 8 = 2,048\end{aligned}$$

$$\begin{aligned}\text{improvement} &= (65,536 - 1,024) / 65,536 \times 100 \\ &= 64,512 / 65,536 \times 100 \\ &= 98.44 \%\end{aligned}$$

Radix-2: 98.44% fewer multiplications – a 64-fold speed-up

Split-radix FFT. It mixes a radix-2 split on the even outputs with a radix-4 split on the odd ones, which is the fewest operations of any power-of-two algorithm:

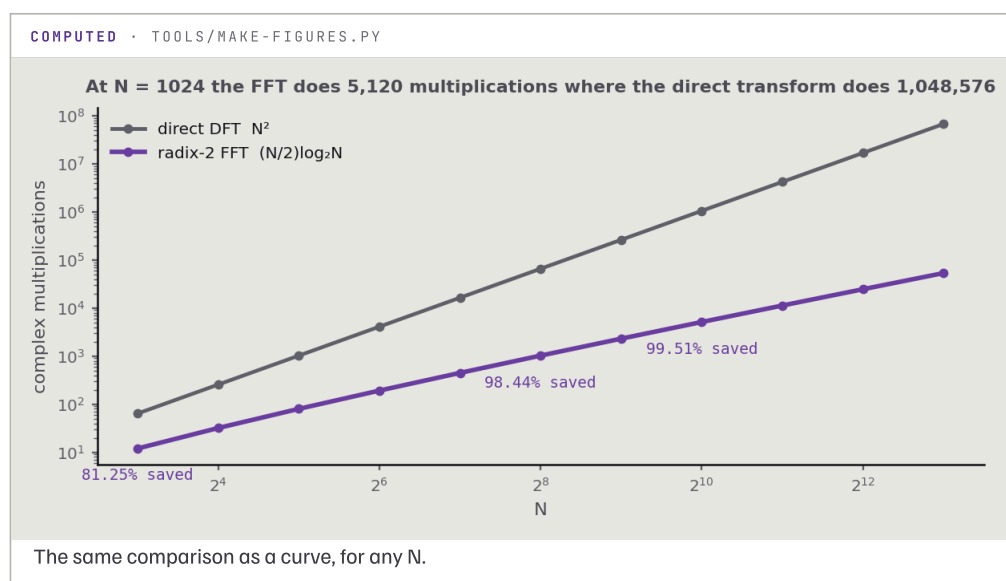
$$\begin{aligned}\text{complex multiplications} &\sim (N/3) \log_2 N = (256/3) \times 8 = 683 \\ \text{improvement} &= (65,536 - 683) / 65,536 \times 100 = 98.96 \%\end{aligned}$$

Split-radix: 98.96% – about a 96-fold speed-up

Put the three side by side, which is what the question is really asking for:

ALGORITHM	COMPLEX MULTIPLICATIONS	SPEED-UP OVER DIRECT
Direct DFT	65,536	—
Radix-2 FFT	1,024	64×
Split-radix	≈ 683	≈ 96×

Split-radix beats radix-2 by roughly a further third, which sounds small against 64× — but on a processor doing this millions of times a second it is worth having.



Q-2 c Explain the Decimation-in-time radix-2 FFT algorithm for calculation of the Inverse Discrete Fourier Transform. Draw the butterfly diagram with a detailed explanation of each step. How many multipliers

04

and adders are needed?

SOLUTION

The point of the question. The IDFT differs from the DFT in only two ways:

$$\begin{aligned} \text{DFT : } X(k) &= \sum x(n) W_N^{+nk} \\ \text{IDFT : } x(n) &= \left(\frac{1}{N}\right) \sum X(k) W_N^{-nk} \end{aligned}$$

So the twiddle exponent changes sign and there is a $1/N$ at the end. **The butterfly structure is completely unchanged** — only the twiddle values are conjugated.

Two ways to do it:

1. **Change the twiddles.** Replace every W^r with W^{-r} (its conjugate) and scale the final output by $1/N$. Same diagram, same operation count.
2. **Reuse the forward FFT unchanged.** Since conjugating flips the sign of the exponent:

$$x(n) = \left(\frac{1}{N}\right) \cdot \text{conj}[\text{FFT}(\text{conj}(X(k)))]$$

Conjugate the input, run the ordinary forward FFT, conjugate the result, divide by N . One hardware block then serves both directions, which is why a real FFT chip has no separate inverse unit.

The butterfly (DIT, inverse):

$$\begin{array}{lcl} a & \text{-----} o \text{-----} o \text{-----} & \rightarrow a + W^{-r} b \\ & \backslash \quad \quad / & \\ b & \text{---} [W^{-r}] \text{---} o \text{-----} o \text{-----} & \rightarrow a - W^{-r} b \end{array}$$

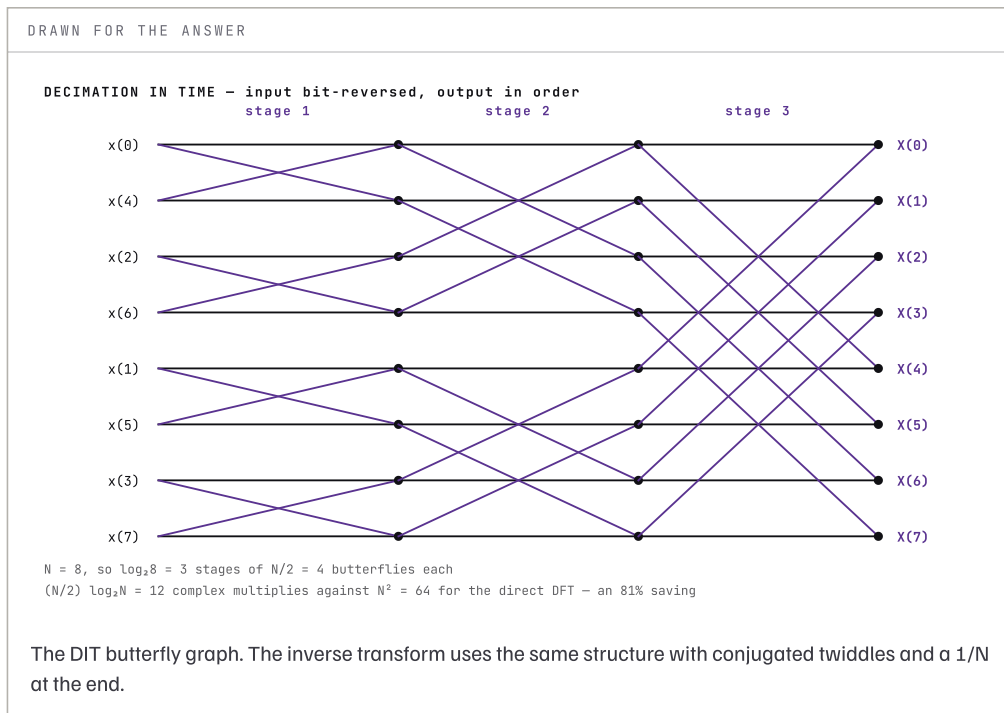
Ordering: DIT means the input $X(k)$ goes in **bit-reversed** and $x(n)$ comes out in **natural** order — same as the forward DIT.

Counts for an N -point inverse FFT:

$$\begin{aligned} \text{stages} &= \log_2 N \\ \text{butterflies} &= \left(\frac{N}{2}\right) \log_2 N \\ \text{complex multipliers} &= \left(\frac{N}{2}\right) \log_2 N && \text{one per butterfly} \\ \text{complex adders} &= N \log_2 N \text{ (two per butterfly)} \\ \text{plus } N \text{ divisions by } N \text{ at the end} \\ \text{for } N = 8: & \quad 12 \text{ multipliers, } 24 \text{ adders, } 8 \text{ scalings} \end{aligned}$$

THE LINE THAT EARNS THE MARK

The inverse FFT costs exactly the same as the forward FFT, apart from the final scaling by $1/N$. If you say only that plus the conjugate trick, you have answered the question.



Q-3 a Determine the particular solution of the difference equation $y(n) - \frac{1}{2}y(n-1) = x(n)$, when the forcing function is $x(n) = 2^n \cdot u(n)$.

02

SOLUTION

For a forcing function of the form a^n , try a particular solution of the **same form**: $y_p(n) = K \cdot 2^n$ for $n \geq 0$.

$$K \cdot 2^n - \left(\frac{1}{2}\right) K \cdot 2^{n-1} = 2^n$$

note $2^{n-1} = 2^n / 2$, so the second term is $(1/4) K \cdot 2^n$:

$$K \cdot 2^n - \left(\frac{1}{4}\right) K \cdot 2^n = 2^n$$

divide through by 2^n :

$$K \left(1 - \frac{1}{4}\right) = 1$$

$$K \left(\frac{3}{4}\right) = 1$$

$$K = \frac{4}{3}$$

$$y_p(n) = \left(\frac{4}{3}\right) \cdot 2^n, \text{ for } n \geq 0$$

Note what the question did and did not ask. Only the particular solution. The *complete* solution would add the homogeneous part $C(\frac{1}{2})^n$, with C fixed by the initial condition — but adding it here would be answering a question that was not set.

WHEN THE TRIAL FORM FAILS

If the forcing function had been $(\frac{1}{2})^n$ — the same as the natural mode — then $K(\frac{1}{2})^n$ would give $0 = (\frac{1}{2})^n$, which is impossible. In that case multiply the trial by n and use $Kn(\frac{1}{2})^n$ instead. Check for that clash before you start.

- Q-3 b** For the given RC filter circuit: (i) calculate the transfer function, (ii) identify whether the circuit is low-pass, high-pass, band-pass or another type, (iii) what is the cut-off frequency of the circuit?

04

SOLUTION

The circuit in the scan is a series R with C to ground, with the output taken across the capacitor
— the standard first-order RC low-pass.

(i) **Transfer function** by the potential divider rule, with the capacitor's impedance $1/(sC)$:

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{\frac{1}{sC}}{R + \frac{1}{sC}}$$

multiply top and bottom by sC :

$$H(s) = \frac{1}{1 + sRC}$$

In the frequency domain, $s = j\omega$:

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\angle H(j\omega) = -\arctan(\omega RC)$$

(ii) **Which type?** Check the two extremes:

$$\begin{aligned} \omega \rightarrow 0 & : |H| \rightarrow 1 & (\text{DC passes untouched}) \\ \omega \rightarrow \infty & : |H| \rightarrow 0 & (\text{high frequencies are killed}) \end{aligned}$$

First-order low-pass filter

If the output were taken across the *resistor* instead, $H(s) = sRC/(1+sRC)$ and it would be a **high-pass**. The component placement is what decides it, not the components themselves.

(iii) **Cut-off frequency** — where the gain falls to $1/\sqrt{2}$ of its maximum, which is -3 dB:

$$\begin{aligned} \frac{1}{\sqrt{1 + (\omega RC)^2}} &= \frac{1}{\sqrt{2}} \\ 1 + (\omega RC)^2 &= 2 \\ (\omega RC)^2 &= 1 \\ \omega RC &= 1 \\ \omega_c &= \frac{1}{RC} \text{ rad/s} \quad f_c = \frac{1}{2\pi RC} \text{ Hz} \end{aligned}$$

At that frequency the phase is $-\arctan(1) = -45^\circ$, which is the other signature of a first-order cut-off.

- Q-4 a** The system function is realised using the input–output relation $2y(n) + y(n-2) - 3y(n-3) = x(n) - x(n-1) + 2x(n-2) - 3x(n-4)$. Realise its structure using delay blocks, adders and multipliers. What is the overall DC gain? Find the frequency response and comment on whether the phase is linear.

03

SOLUTION

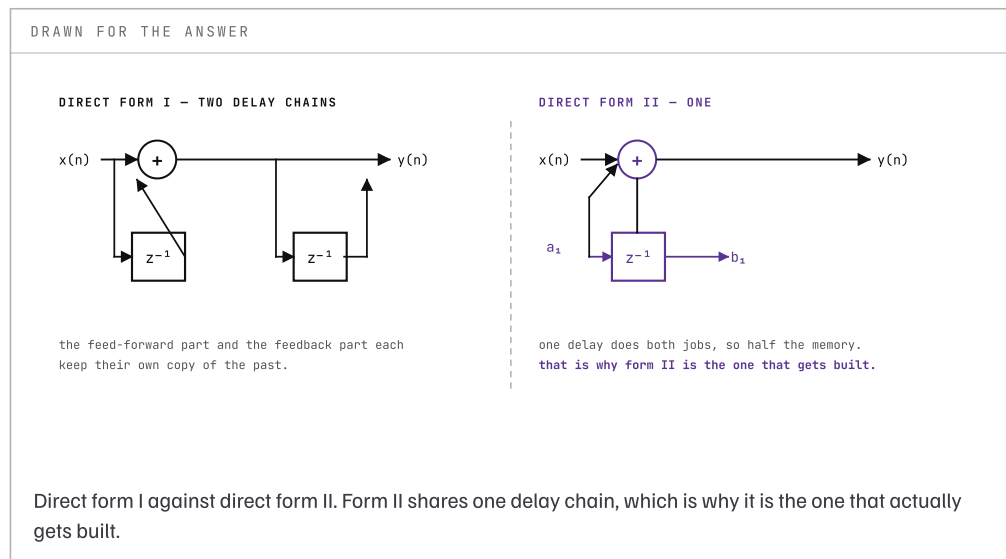
Step 1 — normalise so the $y(n)$ coefficient is 1. Divide everything by 2:

$$y(n) = -0.5y(n-2) + 1.5y(n-3) + 0.5x(n) - 0.5x(n-1) + x(n-2) - 1.5x(n-4)$$

Step 2 — transfer function.

$$H(z) = \frac{0.5 - 0.5z^{-1} + z^{-2} - 1.5z^{-4}}{1 + 0.5z^{-2} - 1.5z^{-3}}$$

Step 3 — Direct Form II structure. Because the same delay chain can serve both halves, Direct Form II needs only $\max(4, 3) = 4$ delay elements, not 7.



DC gain — set $z = 1$, which means summing the coefficients:

$$\begin{aligned} H(1) &= \frac{0.5 - 0.5 + 1 - 1.5}{1 + 0.5 - 1.5} \\ &= \frac{-0.5}{0} \\ &= \text{infinite} \end{aligned}$$

The DC gain is infinite — the denominator has a root at $z = 1$

That is worth saying plainly: $1 + 0.5z^{-2} - 1.5z^{-3}$ vanishes at $z = 1$, so there is a **pole exactly on the unit circle** at DC. The system is therefore **not BIBO stable**, and it behaves like an accumulator at low frequencies.

Frequency response — substitute $z = e^{j\omega}$:

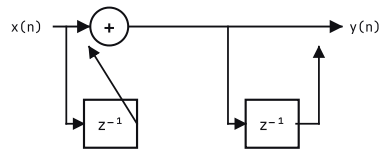
$$\begin{aligned} H(e^{j\omega}) &= (0.5 - 0.5e^{-j\omega} + e^{-j2\omega} - 1.5e^{-j4\omega}) \\ &\quad / (1 + 0.5e^{-j2\omega} - 1.5e^{-j3\omega}) \end{aligned}$$

Is the phase linear? No. Linear phase requires the impulse response to be symmetric or antisymmetric about its centre, and that can only happen in an **FIR** system. This one has feedback, so it is **IIR** with an infinite impulse response and no centre of symmetry. Even the

numerator alone is not symmetric: the coefficients read 0.5, -0.5, 1, 0, -1.5, which is neither a palindrome nor its negative.

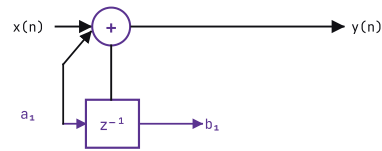
DRAWN FOR THE ANSWER

DIRECT FORM I - TWO DELAY CHAINS



the feed-forward part and the feedback part each keep their own copy of the past.

DIRECT FORM II - ONE



one delay does both jobs, so half the memory.
that is why form II is the one that gets built.

Direct form I against direct form II. Form II shares one delay chain, which is why it is the one that gets built.

MID-SEM, SEPTEMBER 2023

30 MARKS · 25 SEPT 2023

Four questions with internal choice. This scan is the worst of the three — several coefficients are simply illegible — so the questions below are reconstructed only as far as they can be read honestly, and the solutions concentrate on the method.

SVNIT, Surat

Department of Electronics Engineering
Mid-Semester Examination, September 2023
B.Tech. III Sem. V · 25/09/2023

DIGITAL SIGNAL PROCESSING

TOTAL MARKS: 30

DURATION: 90 MINUTES

ATTEMPT ALL QUESTIONS

- Q-1 a** Determine whether each of the following signals is periodic, and if so give its fundamental period. Also say whether each is an energy signal or a power signal. (1) $x(n) = \cos(\pi n/8 + 0.2)$ (2) $x(n) = e^{jn} \cdot \cos(n)$. 04

SOLUTION

The periodicity test. A discrete sinusoid $\cos(\omega_0 n + \phi)$ is periodic only if $\omega_0/2\pi$ is a **rational number**. Then $N = 2\pi k/\omega_0$ for the smallest integer k that makes N a whole number.

(1) $x(n) = \cos(\pi n/8 + 0.2)$

$$\omega_0 = \frac{\pi}{8}$$

$$N = \frac{2\pi k}{\omega_0} = \frac{2\pi k}{\frac{\pi}{8}} = 16k$$

smallest whole number : $k = 1 \rightarrow N = 16$

Periodic, fundamental period $N = 16$

The phase 0.2 shifts the wave along but cannot change its period.

Energy or power? A sinusoid never dies away, so

$$E = \sum |x(n)|^2 \rightarrow \text{infinite}$$

$$P = \text{average of } |x(n)|^2 \text{ over a period} = \frac{1}{2} (\text{finite, non-zero})$$

Power signal, with average power $\frac{1}{2}$

(2) $x(n) = e^{jn} \cdot \cos(n)$

both factors have $\omega_0 = 1$

$$N = \frac{2\pi k}{1} = 2\pi k$$

$2\pi k$ is never a whole number for any integer k , because π is irrational.

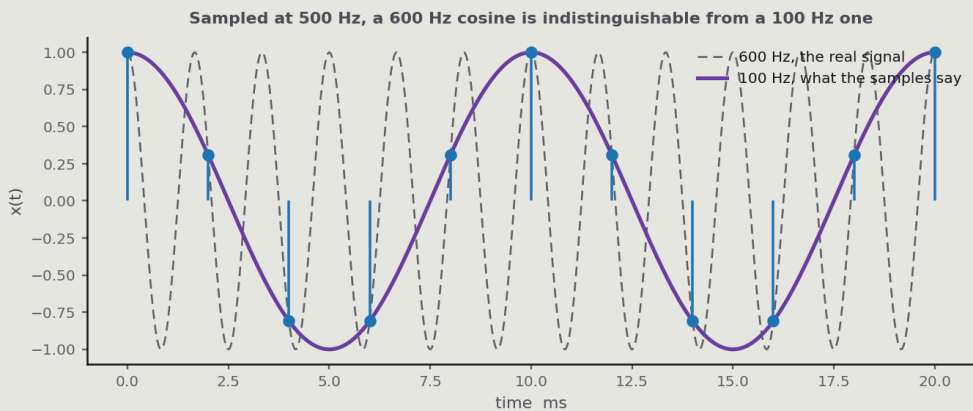
Not periodic

Its magnitude is $|e^{jn} \cos(n)| = |\cos n|$, which is bounded but does not decay, so the energy is infinite and the average power is finite and non-zero — a **power signal**.

THE IDEA BEHIND THE TEST

The continuous-time signal $\cos(t)$ is perfectly periodic. Sampling it can destroy that, because the samples only line up again if the period contains a whole number of sample intervals. This is one of the genuine differences between the discrete and continuous worlds, and it is a favourite opening question.

COMPUTED · TOOLS/MAKE-FIGURES.PY



Aliasing, simulated: a 600 Hz cosine sampled at 500 Hz passes through exactly the same sample values as a 100 Hz one. Once the samples are taken the difference is gone for good.

Q-1 b Determine the response of a system by the convolution method. The system's impulse response and the input signal are given in the paper; the exponents are not legible in the scan. *The method is worked below on a representative pair.*

03

SOLUTION

Take $h(n) = (\frac{1}{2})^n u(n)$ and $x(n) = (\frac{1}{3})^n u(n)$, the usual shape of this question.

Method 1 — direct convolution sum.

$$\begin{aligned} y(n) &= \sum_{k=0}^n x(k)h(n-k) \\ &= \sum_{k=0}^n \left(\frac{1}{3}\right)^k \left(\frac{1}{2}\right)^{n-k} \\ &= \left(\frac{1}{2}\right)^n \sum_{k=0}^n \left(\frac{1}{3}\right)^k (2)^k \\ &= \left(\frac{1}{2}\right)^n \sum_{k=0}^n \left(\frac{2}{3}\right)^k \end{aligned}$$

The sum is a finite geometric series with ratio $r = 2/3$:

$$\begin{aligned}
 \sum_{k=0}^n r^k &= \frac{1 - r^{n+1}}{1 - r} \\
 &= \frac{1 - \left(\frac{2}{3}\right)^{n+1}}{\frac{1}{3}} \\
 &= 3 \left[1 - \left(\frac{2}{3}\right)^{n+1} \right] \\
 y(n) &= 3 \left(\frac{1}{2}\right)^n \left[1 - \left(\frac{2}{3}\right)^{n+1} \right] u(n)
 \end{aligned}$$

Expand to separate the two modes:

$$\begin{aligned}
 y(n) &= 3 \left(\frac{1}{2}\right)^n - 3 \left(\frac{2}{3}\right) \left(\frac{2}{3}\right)^n \left(\frac{1}{2}\right)^n \\
 &= 3 \left(\frac{1}{2}\right)^n - 2 \left(\frac{1}{3}\right)^n u(n)
 \end{aligned}$$

$$y(n) = [3\left(\frac{1}{2}\right)^n - 2\left(\frac{1}{3}\right)^n] \cdot u(n)$$

Method 2 — the z-transform, as a check.

$$\begin{aligned}
 H(z) &= \frac{1}{1 - 0.5z^{-1}} \quad X(z) = \frac{1}{1 - \left(\frac{1}{3}\right)z^{-1}} \\
 Y(z) &= \frac{1}{\left[(1 - 0.5z^{-1})(1 - \left(\frac{1}{3}\right)z^{-1})\right]} \\
 &= \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - \left(\frac{1}{3}\right)z^{-1}} \\
 A &= \frac{1}{1 - \frac{\frac{1}{3}}{0.5}} = \frac{1}{1 - \frac{2}{3}} = 3 \\
 B &= \frac{1}{1 - \frac{0.5}{\frac{1}{3}}} = \frac{1}{1 - 1.5} = -2 \\
 y(n) &= 3(0.5)^n - 2\left(\frac{1}{3}\right)^n, n \geq 0 \quad \text{same answer}
 \end{aligned}$$

Check $n = 0$ and $n = 1$ by hand:

$$\begin{aligned}
 y(0) &= x(0)h(0) = 1 \text{ formula: } 3 - 2 = 1 \text{ OK} \\
 y(1) &= x(0)h(1) + x(1)h(0) \\
 &= 1(0.5) + \left(\frac{1}{3}\right)(1) = 0.8333 \text{ formula: } 1.5 - 0.667 = 0.833 \text{ OK}
 \end{aligned}$$

DRAWN FOR THE ANSWER

FLIP, SLIDE, MULTIPLY, ADD

$x(k)$

	-2	-1	0	1	2
				3	4
			1	2	3

$h(0-k)$

2	-1	1			
---	----	---	--	--	--

$y(0) = 1$

$h(1-k)$

	2	-1	1		
--	---	----	---	--	--

$y(1) = 1$

$h(2-k)$

		2	-1	1	
--	--	---	----	---	--

$y(2) = 3$

$y(n) = \sum_k x(k) h(n-k)$ — flip h , slide it to n , multiply where they overlap, add.
 lengths: $3 + 3 - 1 = 5$ output samples — $\{1, 1, 3, 1, 6\}$.
 free check: $(1+2+3)(1-1+2) = 6 \times 2 = 12$, and $1+1+3+1+6 = 12$.

Convolution as flip, slide, multiply and add — and the length check that catches a wrong answer in two seconds.

Q-2 b

An LTI system is characterised by the system function $H(z) = (3 - 4z^{-1}) / (1 - z^{-1} + 0.5z^{-2})$. Specify the region of convergence and find the impulse response $h(n)$ if (1) the system is stable, (2) the system is causal, (3) the system is anti-causal.

03

SOLUTION

Step 1 — find the poles. Set the denominator to zero. Writing it in z :

$$1 - z^{-1} + 0.5 z^{-2} = 0$$

$$\text{multiply by } z^2 : z^2 - z + 0.5 = 0$$

$$\begin{aligned} z &= [1 \pm \sqrt{1 - 2}] / 2 \\ &= [1 \pm j] / 2 \\ &= 0.5 \pm j0.5 \end{aligned}$$

A complex conjugate pair. Their magnitude is what matters:

$$|z| = \sqrt{0.5^2 + 0.5^2} = \sqrt{0.5} = 0.707$$

Both poles are inside the unit circle — and that single fact answers most of the question.

Step 2 — the three cases. Two poles at the same radius means only **two** possible ROCs, not three.

CASE	ROC	$H(N)$
(2) Causal	$ z > 0.707$ — outside the poles	Right-sided, a decaying oscillation of the form $r^n(A \cos \omega_0 n + B \sin \omega_0 n)u(n)$ with $r = 0.707$ and $\omega_0 = \pi/4$
(1) Stable	$ z > 0.707$ — the same region	The same $h(n)$. Because both poles are inside the unit circle, that ROC already contains $ z = 1$
(3) Anti-causal	$ z < 0.707$ — inside the poles	Left-sided: the same modes but multiplied by $-u(-n-1)$, and not stable, since this ROC excludes the unit circle

The angle of the poles, which sets the oscillation frequency:

$$z = 0.5 + 0.5j = 0.707e^{j\frac{\pi}{4}}$$

so $r = 0.707$ and $\omega_0 = \frac{\pi}{4} \frac{\text{radians}}{\text{sample}}$

Causal and stable are the same ROC here, because every pole is inside the unit circle

THE SENTENCE THE EXAMINER IS WAITING FOR

“Since all poles lie inside the unit circle, the causal solution is also the stable one — so cases (1) and (2) give the identical ROC and the identical $h(n)$.” The question is set to see whether you know that causal and stable coincide exactly when every pole is inside $|z| = 1$.

Q-4 b Suppose we would like to find the N -point DFT where $N = 3^V$. Develop a radix-3 decimation-in-time FFT algorithm, and draw the corresponding flow graph for $N = 9$. How many multiplications are required? Can the computation be performed in place?

04

SOLUTION

The split. Radix-2 divides the input into two interleaved sets; radix-3 divides it into **three**, taken every third sample:

$$\begin{aligned} x(3r) & \quad r = 0 \dots N/3 - 1 \\ x(3r + 1) \\ x(3r + 2) \end{aligned}$$

Each subset gets an $(N/3)$ -point DFT, and the three are recombined with twiddle factors:

$$X(k) = G_0(k) + W_N^k G_1(k) + W_N^{2k} G_2(k)$$

for $k = 0 \dots N/3 - 1$, and then with $W_N^{(N/3)}$ and $W_N^{(2N/3)}$ rotations for the other two thirds of the output

For $N = 9$:

$$\begin{aligned} \text{stages} &= \log_3(9) = 2 \\ \frac{\text{butterflies}}{\text{stage}} &= \frac{N}{3} = 3 \\ \text{each butterfly} &= a3 - \text{point DFT}(3 \text{ inputs}, 3 \text{ outputs}) \end{aligned}$$

$$\begin{aligned} x(0) & \text{ o---} \backslash & / \text{---o } X(0) \\ x(3) & \text{ o-----} [3\text{-pt DFT}] \text{---} [W] \text{---+-----o } X(1) \\ x(6) & \text{ o---} / & \backslash \text{---o } X(2) \end{aligned}$$

$$\begin{aligned} x(1) & \text{ o---} \backslash & / \text{---o } X(3) \\ x(4) & \text{ o-----} [3\text{-pt DFT}] \text{---} [W] \text{---+-----o } X(4) \\ x(7) & \text{ o---} / & \backslash \text{---o } X(5) \end{aligned}$$

$$\begin{aligned} x(2) & \text{ o---} \backslash & / \text{---o } X(6) \\ x(5) & \text{ o-----} [3\text{-pt DFT}] \text{---} [W] \text{---+-----o } X(7) \\ x(8) & \text{ o---} / & \backslash \text{---o } X(8) \end{aligned}$$

stage 1 stage 2
(three 3-point DFTs) (recombine with twiddles)

Input ordering: digit reversal in base 3. Write each index in base 3 and reverse the digits — the base-3 version of bit reversal:

$$\begin{aligned} 0 &= 00 \rightarrow 00 = 0 \cdot 3 = 10 \rightarrow 01 = 1 \cdot 6 = 20 \rightarrow 02 = 2 \\ 1 &= 01 \rightarrow 10 = 3 \cdot 4 = 11 \rightarrow 11 = 4 \cdot 7 = 21 \rightarrow 12 = 5 \\ 2 &= 02 \rightarrow 20 = 6 \cdot 5 = 12 \rightarrow 21 = 7 \cdot 8 = 22 \rightarrow 22 = 8 \end{aligned}$$

input order : $x(0)x(3)x(6)x(1)x(4)x(7)x(2)x(5)x(8)$

Multiplication count. Generalising the radix-2 formula $(N/2)\log_2 N$ to radix r :

complex multiplications $\sim N (r - 1) / r \times \log_r N$

for $N = 9$, $r = 3$:
 $= 9 \times (2/3) \times 2 = 12$ complex multiplications

against the direct DFT: $N^2 = 81$
 speed-up ~ 6.75 times

Can it be done in place? Yes. Each radix-3 butterfly reads three values, computes three outputs, and writes them straight back over the same three locations. Nothing else in the array is touched, so no second array is needed — exactly as with radix-2. The digit-reversal shuffle is the price paid for it.

WHY RADIX-3 EXISTS AT ALL

Radix-2 only works when N is a power of two. If your data length is naturally 9, 27 or 81, zero-padding up to 16, 32 or 128 changes the frequency grid. Radix-3 lets you keep the natural length. The same divide-and-conquer argument drives every radix — split the DFT into smaller DFTs and reuse the twiddle symmetry.